

Homework 4

Dynamic Response (Chapter 3)

- Problems: # 26, 27, 28, 30, 31, 33

Nonlinear Systems (Chapter 9)

- Problem # 1. Use Matlab to help with the calculations if needed. Note the equation of motion in 9.1b should be $\ddot{\theta} + \frac{g}{l}\theta = 0$.
- Problem # 2. Note that $g(u-v)$ is a function evaluated at $u-v$, where $g(x) = x(x-1)(x-4)$ as defined in the problem. Use Matlab to help with the calculations if needed. For example, the following code takes the derivative of f with respect to x then evaluates the result at $x = a$.

```
syms x
a = 0

f = (1-x)*(-x)*(-x-3)
df = diff(f,x)
df_at_a = subs(df,x,a)
```

Optional Bonus Problems

- Consider $1 \times N$ vectors $t[k]$ and $y[k]$ representing time and the step response of a dynamic system, where k is the time index. Write a MATLAB function that takes the vectors as inputs and automatically calculates the rise time, 2% settling time, and percent overshoot.

$$[Tr, Ts, Mp] = \text{sysDynamics}(t, y)$$

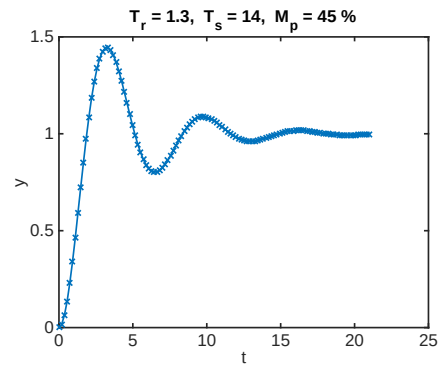
Assume

- At $t[1]$, the output $y[1]$ is the initial steady-state value. The step input (not necessarily unit step) is applied at this time.
- $y[N]$ is the final steady-state value.

Test your code with the following

```
G = tf(1, [1 .5 1]);
[y,t] = step(G);
[Tr, Ts, Mp] = sysDynamics(t, y)
plot(t, y, 'x-')
```

Sample output is below (exact values will vary based on numerical approximations).



2. Extend your code by varying the damping and plotting T_r , T_s , and M_p as a function of damping. For example, you can use a loop like follows:

```
b = linspace(.1,2,100);
for k= 1:length(b)
    G = tf(1, [1 2*b(k) 1]);
    [y,t] = step(G);
    [Tr(k), Ts(k), Mp(k)] = sysDynamics(t, y);
end
```

Briefly explain your output. Example plots are shown below:

